

D464

Managing Operations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A furniture plant, recently certified to ISO 9001, produced 4,800 chairs last month using 1,600 labor hours. The plant manager has been there nine years. This month it produced 5,000 chairs using 2,000 labor hours. What happened to labor productivity?

- A. It fell from 3.0 to 2.5 chairs per labor hour.
- B. It rose, because output rose by 200 chairs.
- C. It fell from 3.0 to 2.4 chairs per labor hour.
- D. It rose from 0.33 to 0.40 hours per chair.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A claims office must process 1,500 claims a week at a productivity of 12 claims per staff hour. How many staff hours does it need?

- A. 125 hours
- B. 18,000 hours
- C. 12 hours
- D. It cannot be found without wage rates.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A bottling plant has a design capacity of 10,000 cases per day and an effective capacity of 8,000 after planned maintenance and changeovers. It was built in 1998 and runs two shifts. Actual output yesterday was 7,200 cases. What is its efficiency?

- A. 72%
- B. 90%
- C. 111%
- D. 80%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A call center's design capacity is 1,200 calls per day, and its utilization yesterday was 75%. How many calls did it handle?

- A.** 300
- B.** 900
- C.** It cannot be found without effective capacity.
- D.** 1,600

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A sandwich shop line has three stations in sequence: order-taking handles 60 customers per hour, assembly 40 per hour, and payment 50 per hour. The shop is known for its sourdough and plans a second location next year. What is the most customers the line can serve per hour?

- A.** 150
- B.** 60
- C.** 40
- D.** 50

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A two-step process has one worker at each step. Step 1 takes 4 minutes per unit and step 2 takes 6 minutes per unit. What is the maximum output in an 8-hour shift?

- A.** 120 units
- B.** 48 units
- C.** 200 units
- D.** 80 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A loan office, which moved to paperless files last year, approves an average of 20 applications per day. On average 140 applications are in process. Its 12 underwriters each hold a professional license. How long does an application spend in the process on average?

- A. 2,800 days
- B. 20 days
- C. About 11.7 days
- D. 7 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A warehouse ships 300 orders per day and targets an average order flow time of 2 days. How many orders should it expect in process on average?

- A. 150
- B. 600
- C. 300
- D. 0, because lean warehouses hold no work in process.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A company makes custom industrial valves; each order has unique specs, low volume, and its own routing through the shop. The company has grown to 250 employees and exports to six countries. Which process type fits best?

- A. Job shop.
- B. Continuous process.
- C. Batch process, because the company is now mid-sized.
- D. Repetitive assembly line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A distribution center team, led by a supervisor who won last year's safety award, piloted a new pick-path layout in one aisle and recorded walking times for two weeks. What is the next PDCA step before changing all 30 aisles?

- A.** Act: roll the layout out to all 30 aisles now.
- B.** Stop; improvement ends once a pilot is run.
- C.** Check: compare the pilot's walking times to the baseline and goal and look for side effects.
- D.** Plan a new, unrelated project.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A packaging firm, publicly traded since 2004, has adopted a triple-bottom-line strategy. It is choosing between Supplier X, which costs 3% less, and Supplier Y, which costs slightly more but uses 60% recycled material and cuts delivery emissions in half. Which approach is most consistent with its strategy?

- A.** Choose Y regardless of how large the cost gap becomes.
- B.** Report emissions data to investors and keep buying on price.
- C.** Choose X, because the lowest price always wins.
- D.** Compare both suppliers on economic, environmental, and social results together; Y may win despite its higher price.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A consumer electronics firm with a well-known brand learns from an audit that a key subcontractor, which offers the lowest price in the region, is holding workers' passports. Switching suppliers would delay a product launch by six weeks. What is the most appropriate operations response?

- A.** Have the subcontractor sign a code of conduct and take no further action.
- B.** Keep using the subcontractor until after launch, then address it.
- C.** Require immediate, verified remediation or move the work, accepting the launch delay if needed.
- D.** Keep the finding confidential to protect the brand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A furniture plant, ~~recently certified to ISO 9001~~, produced 4,800 chairs last month using 1,600 labor hours. ~~The plant manager has been there nine years.~~ This month it produced 5,000 chairs using 2,000 labor hours. What happened to labor productivity?

Target word: change in labor productivity **Decides:** Chairs per labor hour: 4,800/1,600 vs. 5,000/2,000.

A	Correct	$4,800 / 1,600 = 3.0$; $5,000 / 2,000 = 2.5$.
B	Dropped element	Ignores the 25% jump in labor input.
C	Wrong stem number	$4,800 / 2,000$ mixes last month's output with this month's hours.
D	Inverted operation	Hours per chair going up means productivity went down.

2. Answer A Concept 1 • transfer

A claims office must process 1,500 claims a week at a productivity of 12 claims per staff hour. How many staff hours does it need?

Target word: solve for input **Decides:** Input = output / productivity.

A	Correct	$1,500 / 12 = 125$.
B	Inverted operation	Multiplies output by productivity.
C	Wrong stem number	The productivity rate, not the hours needed.
D	Adjacent category	Wages matter for labor cost, not for hours needed.

3. Answer B Concept 2

A bottling plant has a design capacity of 10,000 cases per day and an effective capacity of 8,000 after planned maintenance and changeovers. ~~It was built in 1998 and runs two shifts.~~ Actual output yesterday was 7,200 cases. What is its efficiency?

Target word: efficiency **Decides:** Efficiency uses effective capacity: 7,200 / 8,000.

A	Confusion set	$7,200 / 10,000$ is utilization, not efficiency.
B	Correct	$7,200 / 8,000 = 90\%$.
C	Inverted operation	$8,000 / 7,200$ divides the wrong way.
D	Wrong stem number	$8,000 / 10,000$ compares the two capacities.

4. Answer B Concept 2 • transfer

A call center's design capacity is 1,200 calls per day, and its utilization yesterday was 75%. How many calls did it handle?

Target word: solve for actual output **Decides:** Actual = utilization × design capacity.

A	Confusion set	That is the unused capacity.
B	Correct	$0.75 \times 1,200 = 900$.
C	Adjacent category	Effective capacity is needed for efficiency, not utilization.
D	Inverted operation	Divides 1,200 by 0.75.

5. Answer C Concept 3

A sandwich shop line has three stations in sequence: order-taking handles 60 customers per hour, assembly 40 per hour, and payment 50 per hour. ~~The shop is known for its sourdough and plans a second location next year.~~ What is the most customers the line can serve per hour?

Target word: maximum throughput **Decides:** Assembly is the slowest station.

A	Inverted operation	Adds rates of stations that work in sequence.
B	Wrong stem number	The first station's rate.
C	Correct	The bottleneck (assembly) caps the whole line.
D	Confusion set	The average of the three rates; the line cannot beat its slowest step.

6. Answer D Concept 3 • transfer

A two-step process has one worker at each step. Step 1 takes 4 minutes per unit and step 2 takes 6 minutes per unit. What is the maximum output in an 8-hour shift?

Target word: output from bottleneck time **Decides:** Bottleneck is 6 minutes per unit; $480 / 6$.

A	Wrong stem number	Uses step 1's 4 minutes.
B	Adjacent category	$480 / 10$ treats total flow time as the pace; the steps work in parallel on different units.
C	Inverted operation	Adds the two steps' outputs ($120 + 80$).
D	Correct	$480 \text{ minutes} / 6 \text{ minutes} = 80$.

7. Answer D Concept 4

A loan office, ~~which moved to paperless files last year,~~ approves an average of 20 applications per day. On average 140 applications are in process. ~~Its 12 underwriters each hold a professional license.~~ How long does an application spend in the process on average?

Target word: flow time **Decides:** Flow time = WIP / throughput.

A	Inverted operation	Multiplies WIP by throughput.
B	Wrong stem number	The throughput rate, not the time.
C	Camouflage	Divides by the 12 underwriters, which do not enter Little's Law.
D	Correct	$140 / 20 = 7 \text{ days}$.

8. Answer B Concept 4 • transfer

A warehouse ships 300 orders per day and targets an average order flow time of 2 days. How many orders should it expect in process on average?

Target word: solve for WIP **Decides:** $\text{WIP} = \text{throughput} \times \text{flow time}$.

A	Inverted operation	Divides throughput by flow time.
B	Correct	$300 \times 2 = 600$.
C	Wrong stem number	The daily rate, not the WIP.
D	Absolute language	Any process with flow time has some WIP.

9. Answer A Concept 5

A company makes custom industrial valves; each order has unique specs, low volume, and its own routing through the shop. ~~The company has grown to 250 employees and exports to six countries.~~ Which process type fits best?

Target word: process type **Decides:** Low volume, high variety, varying routings.

A	Correct	Flexible, general-purpose resources for unique, low-volume orders.
B	Inverted operation	The opposite end: very high volume, one standard product.
C	Camouflage	Company size does not set process type; volume and variety do.
D	Adjacent category	Fits high-volume standard products with a fixed sequence.

10. Answer C Concept 6

A distribution center team, ~~led by a supervisor who won last year's safety award,~~ piloted a new pick-path layout in one aisle and recorded walking times for two weeks. What is the next PDCA step before changing all 30 aisles?

Target word: next PDCA step **Decides:** The pilot (Do) is complete; results have not been compared.

A	Dropped element	Skips Check; the gain is not yet confirmed.
B	Absolute language	PDCA is a repeating cycle.
C	Correct	Check follows Do; results decide whether to standardize.
D	Adjacent category	The current cycle is not finished.

11. Answer D Concept 7

A packaging firm, ~~publicly traded since 2004,~~ has adopted a triple-bottom-line strategy. It is choosing between Supplier X, which costs 3% less, and Supplier Y, which costs slightly more but uses 60% recycled material and cuts delivery emissions in half. Which approach is most consistent with its strategy?

Target word: decision approach **Decides:** Triple bottom line weighs people, planet, and profit together.

A	Dropped element	Drops the profit leg of the triple bottom line.
B	Adjacent category	Disclosure is not the same as factoring impacts into decisions.
C	Absolute language	Ignores two of the three bottom lines.
D	Correct	The triple bottom line balances all three, not cost alone.

12. Answer C Concept 8

A consumer electronics firm ~~with a well-known brand~~ learns from an audit that a key subcontractor, ~~which offers the lowest price in the region,~~ is holding workers' passports. Switching suppliers would delay a product launch by six weeks. What is the most appropriate operations response?

Target word: response to a labor violation **Decides:** Holding passports is a forced-labor indicator.

A	Adjacent category	Paper commitments without verification do not fix the violation.
B	Dropped element	Drops the ethical constraint in favor of schedule.
C	Correct	Forced-labor indicators require verified correction or exit.
D	Camouflage	Brand protection is not a reason to leave the violation in place.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Productivity	Productivity = output / input; compare periods on the same ratio.	Calling rising output a productivity gain when input rose faster.	Output and labor hours for two periods.
2	Utilization vs. efficiency	Utilization = actual output / design capacity; efficiency = actual output / effective capacity.	Using design capacity for efficiency.	Both design and effective capacity in one stem.
3	Bottleneck sets throughput	A process can produce no faster than its slowest step.	Averaging or adding station rates.	A list of station rates or times.
4	Little's Law	Inventory (WIP) = throughput × flow time.	Multiplying when you should divide, or using a resource count.	Two of: items in process, rate per period, time in system.
5	Process types	Low-volume, high-variety custom work fits a job shop; high-volume standard goods fit lines or continuous flow.	Matching process type to company size instead of volume and variety.	Unique specs and routings vs. identical units.
6	PDCA continuous improvement	Plan, Do (pilot), Check results against the baseline, then Act to standardize or adjust.	Rolling out right after a pilot without checking.	A pilot just finished.
7	Triple bottom line	Sustainable operations weigh economic, environmental, and social results together.	Treating sustainability as either irrelevant or overriding cost entirely.	A cheaper option vs. a greener option.
8	Ethical sourcing	Serious labor or human-rights violations in the supply chain require verified remediation or exit, even at a cost to schedule.	Accepting a signed code of conduct as enough.	An audit finding plus schedule or price pressure.

Course scope (public description): Managing Operations examines management systems and processes to improve operating efficiency. In this course, students will be introduced to operations management concepts and will learn how analysis of these systems and processes can improve operating efficiency across the organization. Students will explore ethical and sustainable operations management strategies and will analyze operating processes for continuous improvement.